

HEC-RAS 2D Dynamic River Simulation Guide

Technical Documentation

August 3, 2026

1 Introduction

This document deals with the data preparation with Python for the main inputs of the model, that is Terrain and River Flows.

1.1 Part 1: Creating a DEM from LAZ tools

Elevation data for this example needs to be as accurate as possible, so the most reliable source are the Lidas LAZ files downloaded from the National Map: <https://apps.nationalmap.gov/downloader/>

On a Ubuntu running on WSL install pdal

Linux Terminal Session

```
1
2 # Install conda
3 wget https://repo.anaconda.com/miniconda/Miniconda3-latest-Linux-x86_64.
   sh -O ~/miniconda.sh
4
5 # Run the executable
6 bash ~/miniconda.sh
7
8 # Remove the downloaded files
9 rm ~/miniconda.sh
10
11 # Activate conda
12 source ~/.bashrc
13
14 # Check conda version
15 conda --version
16
17 # Install Pdal under a enviroment with conda. It is better to prevent
   package conflict.
18 conda create --name myenv --channel conda-forge pdal gdal
19
20 # List all your enviroments
21 conda env list
22
23 # Activate your enviroment
24 conda activate myenv
25
26 # Check if PDAL is installed
27 pdal --version
28
29 # Check if GDAL is installed
30 gdal --version
```

Once Pdal is installed we are going to create 4 dems from the 4 laz files with scripts like these:

Linux Terminal Session

```
1 pdal translate Laz/USGS_LPC_OH_Statewide_Phase3_2021_B21_BS13400457.laz
2   l1processedData/Rst/partialDem/BS13400457_16N.tif \
3   reprojection \
4   --filters.reprojection.out_srs="EPSG:32616" \
5   --writers.gdal.resolution=1.0 \
6   --writers.gdal.output_type="mean" \
7   --writers.gdal.data_type="float32"
8
9 pdal translate Laz/USGS_LPC_OH_Statewide_Phase3_2021_B21_BS13400458.laz
10  l1processedData/Rst/partialDem/BS13400458_16N.tif \
11  reprojection \
12  --filters.reprojection.out_srs="EPSG:32616" \
13  --writers.gdal.resolution=1.0 \
14  --writers.gdal.output_type="mean" \
15  --writers.gdal.data_type="float32"
16
17 pdal translate Laz/USGS_LPC_OH_Statewide_Phase3_2021_B21_BS13410457.laz
18  l1processedData/Rst/partialDem/BS13410457_16N.tif \
19  reprojection \
20  --filters.reprojection.out_srs="EPSG:32616" \
21  --writers.gdal.resolution=1.0 \
22  --writers.gdal.output_type="mean" \
23  --writers.gdal.data_type="float32"
24
25 pdal translate Laz/USGS_LPC_OH_Statewide_Phase3_2021_B21_BS13410458.laz
26  l1processedData/Rst/partialDem/BS13410458_16N.tif \
27  reprojection \
28  --filters.reprojection.out_srs="EPSG:32616" \
29  --writers.gdal.resolution=1.0 \
30  --writers.gdal.output_type="mean" \
31  --writers.gdal.data_type="float32"
32
33 pdal translate Laz/USGS_LPC_OH_Statewide_Phase3_2021_B21_BS13420457.laz
34  l1processedData/Rst/partialDem/BS13420457_16N.tif \
35  reprojection \
36  --filters.reprojection.out_srs="EPSG:32616" \
37  --writers.gdal.resolution=1.0 \
38  --writers.gdal.output_type="mean" \
39  --writers.gdal.data_type="float32"
40
41 pdal translate Laz/USGS_LPC_OH_Statewide_Phase3_2021_B21_BS13420458.laz
42  l1processedData/Rst/partialDem/BS13420458_16N.tif \
43  reprojection \
44  --filters.reprojection.out_srs="EPSG:32616" \
45  --writers.gdal.resolution=1.0 \
46  --writers.gdal.output_type="mean" \
47  --writers.gdal.data_type="float32"
```

You can check the resulting rasters in QGIS.

Now is time to merge all rasters and clip to model extension extended version to prevent nan values on bordes. We are going to do it with GDAL since it is faster.

Linux Terminal Session

```
1 gdalwarp \
```

```

2  -wo CUTLINE_ALL_TOUCHED=TRUE \
3  -cutline l1processedData/Shp/Model/aoiExtended.shp \
4  -crop_to_cutline \
5  -dstnodata -9999 \
6  l1processedData/Rst/partialDem/BS13400457_16N.tif \
7  l1processedData/Rst/partialDem/BS13400458_16N.tif \
8  l1processedData/Rst/partialDem/BS13410457_16N.tif \
9  l1processedData/Rst/partialDem/BS13410458_16N.tif \
10 l1processedData/Rst/partialDem/BS13420457_16N.tif \
11 l1processedData/Rst/partialDem/BS13420458_16N.tif \
12 l1processedData/Rst/partialDem/OHIO_2021_16N_feet.tif

```

But the dem is still on feet, we have to shift it to meters

Linux Terminal Session

```

1 gdal_translate \
2  -ot Float32 \
3  -scale 0 1 0 0.3048 \
4  -co "COMPRESS=DEFLATE" \
5  l1processedData/Rst/partialDem/OHIO_2021_16N_feet.tif l1processedData/
   Rst/OHIO_2021_16N_m.tif

```

2 Creating the Topobathymetric map

Follow the code in s1topoBathymetry.ipynb